

Reviewer Pack — Minimal Eta-Invariant and Communication Complexity Rank Vari...

Entry 7c2722ea41fe · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Eta-Invariant and Communication Complexity Rank Variance

Entry ID: 7c2722ea41fe **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-08 17:40:48 UTC

1. Conjecture

Field A (mathematical branch): Category Theory (Eta-Invariant) **Field B** (complexity object): Communication Complexity (Matrix Rank Variance)

Statement:

For every Boolean function f with communication complexity rank variance $R(f)$, the minimal eta-invariant of its associated category C_f is linearly correlated with $R(f)$ such that $\eta(C_f) = \Theta(R(f))$.

Rationale (proposer's reasoning):

The eta-invariant captures a categorical structure that may reflect the intrinsic complexity of a Boolean function. If this invariant is correlated with communication complexity rank variance, it could provide insights into the structural properties of complex functions and potentially lead to new algorithms for communication complexity.

Taxonomy category: Category Theory - Communication Complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 00610f21d1d92317

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between minimal eta-invariant $\eta(C_f)$ and communication complexity rank variance $R(f)$ for all instances is ≥ 0.8 AND the p-value from a significance test is ≤ 0.05 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "minimal Eta-Invariant" AND "communication complexity rank variance"
- "category theory" AND "matrix rank variance" - "Boolean function" AND "Eta-Invariant" AND communication complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity_rank_variance(f):
        n = int(math.log2(len(f)))
        rank = 0
        for i in range(n):
            if all(f[j] == f[j + 2**i] for j in range(2**(n - i) - 1)):
                rank += 1
        return rank

    def eta_invariant(f):
        n = int(math.log2(len(f)))
        invariant = 0
        for i in range(n):
            for j in range(2**i):
                if f[j] != f[j + 2**i]:
                    invariant += 1
        return invariant / (n * 2**(n - 1))

    n_values = [5, 10, 15, 20, 30, 40]
    instances_tested = 0
    total_eta = 0.0
    total_variance = 0.0

    for n in n_values:
        for _ in range(5): # Ensure at least 30 instances per seed
```

```

        f = generate_boolean_function(n)
        eta = eta_invariant(f)
        variance = communication_complexity_rank_variance(f)
        total_eta += eta
        total_variance += variance
        instances_tested += 1

mean_eta = total_eta / instances_tested
mean_variance = total_variance / instances_tested
correlation_coefficient = (instances_tested * sum(eta * variance for eta, variance in zip([mean_eta, mean_variance],
        instances_tested * mean_eta * mean_variance) / math.sqrt((instances_tested * sum(variance**2 for variance in zip([mean_eta, mean_variance],
        instances_tested * sum(variance**2 for variance in zip([mean_eta, mean_variance],

return {
    "metric_name": "eta_variance_correlation",
    "metric_value": correlation_coefficient,
    "instances_tested": instances_tested,
    "n_max": max(n_values),
    "conjecture_holds": abs(correlation_coefficient) >= 0.8,
    "counterexample": ""
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means the Pearson correlation coefficient and p-value could not be calculated to verify the conjecture. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16860
2	preregistration	ollama_remote	glm4:latest	0	0	9836
3	novelty	ollama_remote	glm4:latest	0	0	9023
4	novelty	ollama_remote	glm4:latest	0	0	21061
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	32600
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12491
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	53078
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11066
9	judge	ollama_remote	glm4:latest	0	0	54139

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 220154 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in *research/audit/7c2722ea41fe.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/7c2722ea41fe.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/7c2722ea41fe.tar.gz* (if generated)